

a. gemini 3.5 Flash-Lite and GPT-5.6 Terra

b. Accessed them both through their websites

c. [Objective

Write a C program that picks a secret number between 1 and 10.

The user gets up to 3 tries to guess the number.

Requirements

- Use a fixed secret number (for example, 7).
- Use printf to prompt the user and scanf to read their guess.
- Use a while or for loop to let the user guess up to 3 times.
- Use if-else statements to check if the guess is too high, too low, or correct.
- Stop the loop immediately if the user guesses the correct number.
- Print a final win or lose message.

Example Output

text

Guess a number between 1 and 10.

Attempt 1/3. Enter your guess: 5

Too low! Try again.

Attempt 2/3. Enter your guess: 8

Too high! Try again.

Attempt 3/3. Enter your guess: 7

Correct! You win!]

d.

Geminis code:

```
#include <stdio.h>
```

```
int main() {
```

```
    int secret = 7;
```

```
    int guess;
```

```
    int max_tries = 3;
```

```
    int won = 0;
```

```
    printf("Guess a number between 1 and 10.\n");
```

```
    for (int i = 1; i <= max_tries; i++) {
```

```
        printf("Attempt %d/%d. Enter your guess: ", i, max_tries);
```

```
        scanf("%d", &guess);
```

```

    if (guess == secret) {
        printf("Correct! You win!\n");
        won = 1;
        break;
    } else if (guess < secret) {
        if (i < max_tries) {
            printf("Too low! Try again.\n");
        }
    } else {
        if (i < max_tries) {
            printf("Too high! Try again.\n");
        }
    }
}

if (!won) {
    printf("Out of tries! You lose. The secret number was %d.\n", secret);
}

return 0;
}

```

GPTs code:

```

#include <stdio.h>

int main(void) {
    const int secretNumber = 7;
    int guess;
    int attempt;
    int won = 0;

    printf("Guess a number between 1 and 10.\n");

    for (attempt = 1; attempt <= 3; attempt++) {
        printf("Attempt %d/3. Enter your guess: ", attempt);

        if (scanf("%d", &guess) != 1) {
            printf("Invalid input. Please enter a whole number.\n");
            return 1;
        }
    }
}

```

```

    }

    if (guess == secretNumber) {
        printf("Correct! You win!\n");
        won = 1;
        break;
    } else if (guess < secretNumber) {
        printf("Too low! Try again.\n");
    } else {
        printf("Too high! Try again.\n");
    }
}

if (!won) {
    printf("Sorry, you lose. The secret number was %d.\n", secretNumber);
}

return 0;
}

```

e. Both examples of code handle sample inputs correctly and both use for loops which are both time and space efficient. They both did not have any comments for maintainability.